

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
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Section / Batch:	Slot C / 2024	Date:	

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2,Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q141. Closest Pair – Smart City Sensor Distribution

A smart city deploys sensors at $\{(2,3), (4,6), (7,8), (3,4), (9,1), (6,5)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest sensors. Clearly interpret the problem and justify the use of the brute force technique. Compute the distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of pairwise comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in sensor placement optimization.

Q118. Bubble Sort – Mixed Small and Large Elements

Apply the Brute Force technique using Bubble Sort on the dataset $[42, 9, 33, 15, 27, 3]$. Clearly interpret the problem and explain how Bubble Sort gradually places larger elements toward the end of the list. Perform the sorting process step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on mixed values.

Q59. Linear Search – Search in Mixed Dataset

Given the unsorted dataset $[8, 19, 27, 44, 12, 31, 50]$, apply Linear Search to find the element 12. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons made until the element is found. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search for moderate-sized unsorted lists.

Q141. Closest Pair – Smart City Sensor Distribution

```
[5]
✓ 0s
a = [8, 19, 27, 44, 12, 31, 50]
key = 12

for i in range(len(a)):
    if a[i] == key:
        print("Element found at position:", i + 1)
        break
    else:
        print("Element not found")

... Element found at position: 5
```

Q118. Bubble Sort – Mixed Small and Large Elements

```
[3]
✓ 0s
a = [42, 9, 33, 15, 27, 3]

for i in range(len(a) - 1):
    for j in range(len(a) - 1 - i):
        if a[j] > a[j + 1]:
            a[j], a[j + 1] = a[j + 1], a[j]
        print("Pass", i + 1, ":", a)

print("Sorted Array:", a)

... Pass 1 : [9, 33, 15, 27, 3, 42]
    Pass 2 : [9, 15, 27, 3, 33, 42]
    Pass 3 : [9, 15, 3, 27, 33, 42]
    Pass 4 : [9, 3, 15, 27, 33, 42]
    Pass 5 : [3, 9, 15, 27, 33, 42]
    Sorted Array: [3, 9, 15, 27, 33, 42]
```

Q59. Linear Search – Search in Mixed Dataset

```
[2]
import math
points = [(2,3), (4,6), (7,8), (3,4), (9,1), (6,5)]
min_dist = float('inf')
closest = ()
for i in range(len(points)):
    for j in range(i + 1, len(points)):
        d = math.sqrt((points[i][0]-points[j][0])**2 +
                      (points[i][1]-points[j][1])**2)
        print(points[i], points[j], "=", round(d,3))
        if d < min_dist:
            min_dist = d
            closest = (points[i], points[j])
print("\nClosest Pair:", closest)
print("Minimum Distance:", round(min_dist,3))

... (2, 3) (4, 6) = 3.606
(2, 3) (7, 8) = 7.071
(2, 3) (3, 4) = 1.414
(2, 3) (9, 1) = 7.28
(2, 3) (6, 5) = 4.472
(4, 6) (7, 8) = 3.606
(4, 6) (3, 4) = 2.236
(4, 6) (9, 1) = 7.071
(4, 6) (6, 5) = 2.236
(7, 8) (3, 4) = 5.657
(7, 8) (9, 1) = 7.28
(7, 8) (6, 5) = 3.162
(3, 4) (9, 1) = 6.708
(3, 4) (6, 5) = 3.162
(9, 1) (6, 5) = 5.0

Closest Pair: ((2, 3), (3, 4))
Minimum Distance: 1.414
```

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____